# CIE 6002 Matrix Analysis Homework 2

Ziheng Wang (zihengwang3@link.cuhk.edu.cn)

March 2, 2026

Declaration: I used LLM for enhancing language, explaining definitions and checking grammar in this homework.

## 1. Problem 2.1

**1.1)** We prove this inequality holds by finding a contradiction case in 2D linear space:

$$\begin{aligned}\mathcal{R} &= \{(x, y) : y = 2x\},\\ \mathcal{S} &= \{(x, y) : y = x\},\\ \mathcal{T} &= \{(x, y) : y = 0.5x\},\end{aligned}$$

where $(2, 3) \in \mathcal{R} \cap (\mathcal{S} + \mathcal{T})$ but $(2, 3) \notin \mathcal{R} \cap \mathcal{S} + \mathcal{R} \cap \mathcal{T}$.

**1.2)** "$\supset$":

- Since $(\mathcal{R} \cap \mathcal{S}) \subset \mathcal{R}$ and $\mathcal{T} \subset \mathcal{R}$, we have $(\mathcal{R} \cap \mathcal{S}) + \mathcal{T} \subset \mathcal{R}$.
- Since $(\mathcal{R} \cap \mathcal{S}) \subset \mathcal{S}$, we have $(\mathcal{R} \cap \mathcal{S}) + \mathcal{T} \subset \mathcal{S} + \mathcal{T}$.

Therefore, we have $\mathcal{R} \cap (\mathcal{S} + \mathcal{T}) \supset (\mathcal{R} \cap \mathcal{S}) + \mathcal{T}$.

"$\subset$":

For all $x \in \mathcal{R} \cap (\mathcal{S} + \mathcal{T})$, we have $x \in \mathcal{R}$ and $x \in \mathcal{S} + \mathcal{T}$. Since $x \in \mathcal{S} + \mathcal{T}$, there exist $s \in \mathcal{S}$ and $t \in \mathcal{T}$ such that $x = s + t$. Since $x \in \mathcal{R}$ and $t \in \mathcal{T} \subset \mathcal{R}$, we have $s = x - t \in \mathcal{R}$. Therefore, we have $s \in \mathcal{R} \cap \mathcal{S}$ and $t \in \mathcal{T}$, which implies that $x = s + t \in (\mathcal{R} \cap \mathcal{S}) + \mathcal{T}$. Therefore, we have $\mathcal{R} \cap (\mathcal{S} + \mathcal{T}) \subset (\mathcal{R} \cap \mathcal{S}) + \mathcal{T}$.

Overall, we can conclude that $\mathcal{R} \cap (\mathcal{S} + \mathcal{T}) = (\mathcal{R} \cap \mathcal{S}) + \mathcal{T}$.

**1.3)** Based on symmetry, we only need to prove

$$\mathcal{R} \cap (\mathcal{S} + \mathcal{T}) = (\mathcal{R} \cap \mathcal{S}) + (\mathcal{R} \cap \mathcal{T}) \Rightarrow \mathcal{S} \cap (\mathcal{R} + \mathcal{T}) = (\mathcal{S} \cap \mathcal{R}) + (\mathcal{S} \cap \mathcal{T}),$$

then all equations hold.

"$\supset$":

All $x \in (\mathcal{S} \cap \mathcal{R}) + (\mathcal{S} \cap \mathcal{T})$ can be written as $x = u + v$, where $u \in (\mathcal{S} \cap \mathcal{R})$ and $v \in (\mathcal{S} \cap \mathcal{T})$. Since $u \in \mathcal{R}$ and $v \in \mathcal{T}$, we have $x = u + v \in \mathcal{R} + \mathcal{T}$. Since $u \in \mathcal{S}$ and $v \in \mathcal{S}$, we have $x = u + v \in \mathcal{S}$. Therefore, we have $(\mathcal{S} \cap \mathcal{R}) + (\mathcal{S} \cap \mathcal{T}) \subset \mathcal{S} \cap (\mathcal{R} + \mathcal{T})$.

"$\subset$":

All $x \in \mathcal{S} \cap (\mathcal{R} + \mathcal{T})$ can be written as $x = r + t$, where $x \in \mathcal{S}$, $r \in \mathcal{R}$ and $t \in \mathcal{T}$. Consider $r = x - t$, since $x \in \mathcal{S} \in (\mathcal{S} + \mathcal{T})$, $t \in \mathcal{T} \in (\mathcal{S} + \mathcal{T})$, we have $r \in \mathcal{S} + \mathcal{T}$.

Given that

$$\mathcal{R} \cap (\mathcal{S} + \mathcal{T}) = (\mathcal{R} \cap \mathcal{S}) + (\mathcal{R} \cap \mathcal{T}),$$

$r$ can be written as $r = r_s + r_t$, where $r_s \in \mathcal{R} \cap \mathcal{S}$ and $r_t \in \mathcal{R} \cap \mathcal{T}$. Substituting this into the expression $x = r + t$, we have

$$x = r_s + r_t + t,$$

where $r_s \in \mathcal{S} \cap \mathcal{R}$. Here $r_t + t$ is in $\mathcal{T}$ since $r_t \in \mathcal{R} \cap \mathcal{T}$ and $t \in \mathcal{T}$. Since $x \in \mathcal{S}$ and $r_s \in \mathcal{S}$, $r_t + t = x - r_s$ implies $r_t + t \in \mathcal{S}$. Then, $r_t + t \in \mathcal{S} \cap \mathcal{T}$.

Overall, we can write $x$ as a sum of two vectors, where one is in $\mathcal{S} \cap \mathcal{R}$ and the other is in $\mathcal{S} \cap \mathcal{T}$. Therefore, we have $\mathcal{S} \cap (\mathcal{R} + \mathcal{T}) \subset (\mathcal{S} \cap \mathcal{R}) + (\mathcal{S} \cap \mathcal{T})$, and we can conclude that $\mathcal{S} \cap (\mathcal{R} + \mathcal{T}) = (\mathcal{S} \cap \mathcal{R}) + (\mathcal{S} \cap \mathcal{T})$.

## 2. Problem 2.2

$$A(\mathcal{U}_1 + \mathcal{U}_2) = A\mathcal{U}_1 + A\mathcal{U}_2 \tag{1}$$

$$A(\mathcal{U}_1 \cap \mathcal{U}_2) \subset A\mathcal{U}_1 \cap A\mathcal{U}_2 \tag{2}$$

$$A^{-1}(\mathcal{V}_1 + \mathcal{V}_2) \supset A^{-1}\mathcal{V}_1 + A^{-1}\mathcal{V}_2 \tag{3}$$

$$A^{-1}(\mathcal{V}_1 \cap \mathcal{V}_2) = A^{-1}\mathcal{V}_1 \cap A^{-1}\mathcal{V}_2. \tag{4}$$

**(a) Verify (Eq. (1) – Eq. (4)):**

Eq. (1) For all $x \in A(\mathcal{U}_1 + \mathcal{U}_2)$, there exists $u_1 \in \mathcal{U}_1$ and $u_2 \in \mathcal{U}_2$ such that $x = A(u_1 + u_2)$. Since $A$ is linear, we have $x = Au_1 + Au_2$, where $Au_1 \in A\mathcal{U}_1$ and $Au_2 \in A\mathcal{U}_2$. Therefore, we have $x \in A\mathcal{U}_1 + A\mathcal{U}_2$, which implies that $A(\mathcal{U}_1 + \mathcal{U}_2) \subset A\mathcal{U}_1 + A\mathcal{U}_2$.

Similarly, we have $A\mathcal{U}_1 + A\mathcal{U}_2 \subset A(\mathcal{U}_1 + \mathcal{U}_2)$, therefore $A(\mathcal{U}_1 + \mathcal{U}_2) = A\mathcal{U}_1 + A\mathcal{U}_2$.

Eq. (2) For all $x \in A(\mathcal{U}_1 \cap \mathcal{U}_2)$, there exists $u \in \mathcal{U}_1 \cap \mathcal{U}_2$ such that $x = Au$. Since $u \in \mathcal{U}_1$ and $u \in \mathcal{U}_2$, we have $x = Au \in A\mathcal{U}_1$ and $x = Au \in A\mathcal{U}_2$. Therefore, we have $x \in A\mathcal{U}_1 \cap A\mathcal{U}_2$, which implies that $A(\mathcal{U}_1 \cap \mathcal{U}_2) \subset A\mathcal{U}_1 \cap A\mathcal{U}_2$.

Eq. (3) For all $x \in A^{-1}\mathcal{V}_1 + A^{-1}\mathcal{V}_2$, we can write $x = x_1 + x_2$ where $x_1 \in A^{-1}\mathcal{V}_1$ and $x_2 \in A^{-1}\mathcal{V}_2$. Then we have $Ax = Ax_1 + Ax_2 \in \mathcal{V}_1 + \mathcal{V}_2$, which implies that $x \in A^{-1}(\mathcal{V}_1 + \mathcal{V}_2)$, and we have $A^{-1}(\mathcal{V}_1 + \mathcal{V}_2) \supset A^{-1}\mathcal{V}_1 + A^{-1}\mathcal{V}_2$.

Eq. (4) For all $x \in A^{-1}(\mathcal{V}_1 \cap \mathcal{V}_2)$, we have

$$\begin{aligned} &Ax \in \mathcal{V}_1 \text{ and } Ax \in \mathcal{V}_2 \\ \Leftrightarrow &x \in A^{-1}\mathcal{V}_1 \text{ and } x \in A^{-1}\mathcal{V}_2 \\ \Leftrightarrow &x \in A^{-1}\mathcal{V}_1 \cap A^{-1}\mathcal{V}_2 \end{aligned}$$

Since each of the steps above is a logical equivalence (if and only if), we have $A^{-1}(\mathcal{V}_1 \cap \mathcal{V}_2) = A^{-1}\mathcal{V}_1 \cap A^{-1}\mathcal{V}_2$.

**(b) Show that Eq. (2) becomes an equality if and only if $\mathcal{N}(A) \cap (\mathcal{U}_1 + \mathcal{U}_2) = \mathcal{N}(A) \cap \mathcal{U}_1 + \mathcal{N}(A) \cap \mathcal{U}_2$: "$\Rightarrow$":**

Suppose Eq. (2) becomes an equality, meaning $A(\mathcal{U}_1 \cap \mathcal{U}_2) = A\mathcal{U}_1 \cap A\mathcal{U}_2$. We only need to show $\mathcal{N}(A) \cap (\mathcal{U}_1 + \mathcal{U}_2) \subset \mathcal{N}(A) \cap \mathcal{U}_1 + \mathcal{N}(A) \cap \mathcal{U}_2$ because the reverse inclusion $\mathcal{N}(A) \cap \mathcal{U}_1 + \mathcal{N}(A) \cap \mathcal{U}_2 \subset \mathcal{N}(A) \cap (\mathcal{U}_1 + \mathcal{U}_2)$ is true for any linear space (we have proved that in Problem 2.1(1.3)).

Let $x \in \mathcal{N}(A) \cap (\mathcal{U}_1 + \mathcal{U}_2)$. By definition, $Ax = 0$, and there exist $u_1 \in \mathcal{U}_1, u_2 \in \mathcal{U}_2$ such that $x = u_1 + u_2$. We have $0 = Ax = Au_1 + Au_2$, which implies $Au_1 = A(-u_2)$.

Let $y = Au_1 = A(-u_2)$. Since $u_1 \in \mathcal{U}_1$ and $-u_2 \in \mathcal{U}_2$ since $\mathcal{U}_2$ is a subspace, we have $y \in A\mathcal{U}_1$ and $y \in A\mathcal{U}_2$, so $y \in A\mathcal{U}_1 \cap A\mathcal{U}_2$.

Since $A(\mathcal{U}_1 \cap \mathcal{U}_2) = A\mathcal{U}_1 \cap A\mathcal{U}_2$, $y \in A(\mathcal{U}_1 \cap \mathcal{U}_2)$, so there exists $v \in \mathcal{U}_1 \cap \mathcal{U}_2$ such that $y = Av$. This means $Au_1 = Av \implies A(u_1 - v) = 0$, so $u_1 - v \in \mathcal{N}(A)$. Since $u_1 \in \mathcal{U}_1$ and $v \in \mathcal{U}_1$, we also have $u_1 - v \in \mathcal{U}_1$. Thus, $u_1 - v \in \mathcal{N}(A) \cap \mathcal{U}_1$. Similarly, $A(-u_2) = Av \implies A(-u_2 - v) = 0 \implies A(u_2 + v) = 0$, so $u_2 + v \in \mathcal{N}(A)$ and $u_2 + v \in \mathcal{U}_2$. Thus, $u_2 + v \in \mathcal{N}(A) \cap \mathcal{U}_2$.

Rearranging the expression $x = u_1 + u_2$ gives $x = (u_1 - v) + (u_2 + v)$, where $u_1 - v \in \mathcal{N}(A) \cap \mathcal{U}_1$ and $u_2 + v \in \mathcal{N}(A) \cap \mathcal{U}_2$. Therefore, we have $x \in \mathcal{N}(A) \cap \mathcal{U}_1 + \mathcal{N}(A) \cap \mathcal{U}_2$, which implies $\mathcal{N}(A) \cap (\mathcal{U}_1 + \mathcal{U}_2) \subset \mathcal{N}(A) \cap \mathcal{U}_1 + \mathcal{N}(A) \cap \mathcal{U}_2$.

**"$\Leftarrow$":**

Suppose $\mathcal{N}(A) \cap (\mathcal{U}_1 + \mathcal{U}_2) = \mathcal{N}(A) \cap \mathcal{U}_1 + \mathcal{N}(A) \cap \mathcal{U}_2$. We only need to show $A\mathcal{U}_1 \cap A\mathcal{U}_2 \subset A(\mathcal{U}_1 \cap \mathcal{U}_2)$, since the reverse inclusion is already established by Eq. (2).

Let $x \in A\mathcal{U}_1 \cap A\mathcal{U}_2$. Then there exist $u_1 \in \mathcal{U}_1$ and $u_2 \in \mathcal{U}_2$ such that $x = Au_1 = Au_2$. This implies $A(u_1 - u_2) = 0$, so $u_1 - u_2 \in \mathcal{N}(A)$. Also, since $u_1 \in \mathcal{U}_1$ and $-u_2 \in \mathcal{U}_2$, we have $u_1 - u_2 \in \mathcal{U}_1 + \mathcal{U}_2$. Thus, $u_1 - u_2 \in \mathcal{N}(A) \cap (\mathcal{U}_1 + \mathcal{U}_2)$.

By our assumption, there exist $n_1 \in \mathcal{N}(A) \cap \mathcal{U}_1$ and $n_2 \in \mathcal{N}(A) \cap \mathcal{U}_2$ such that $u_1 - u_2 = n_1 + n_2$. Rearranging this gives $u_1 - n_1 = u_2 + n_2$. Let $v = u_1 - n_1 = u_2 + n_2$. Since $u_1 \in \mathcal{U}_1$ and $n_1 \in \mathcal{U}_1$, we have $v \in \mathcal{U}_1$. Since $u_2 \in \mathcal{U}_2$ and $n_2 \in \mathcal{U}_2$, we have $v \in \mathcal{U}_2$. Thus $v \in \mathcal{U}_1 \cap \mathcal{U}_2$.

Now, consider $Av$: $Av = A(u_1 - n_1) = Au_1 - An_1 = Au_1 - 0 = x$ (because $n_1 \in \mathcal{N}(A)$). Since $v \in \mathcal{U}_1 \cap \mathcal{U}_2$ and $Av = x$, we have $x \in A(\mathcal{U}_1 \cap \mathcal{U}_2)$. This implies $A\mathcal{U}_1 \cap A\mathcal{U}_2 \subset A(\mathcal{U}_1 \cap \mathcal{U}_2)$, which makes Eq. (2) an equality.

**(c) Show that Eq.** (3) **becomes an equality if and only if** $\mathcal{R}(A) \cap (\mathcal{V}_1 + \mathcal{V}_2) = \mathcal{R}(A) \cap \mathcal{V}_1 + \mathcal{R}(A) \cap \mathcal{V}_2$:

**"$\Rightarrow$":**

Suppose Eq. (3) becomes an equality, meaning $A^{-1}(\mathcal{V}_1 + \mathcal{V}_2) = A^{-1}\mathcal{V}_1 + A^{-1}\mathcal{V}_2$. We only need to show $\mathcal{R}(A) \cap (\mathcal{V}_1 + \mathcal{V}_2) \subset \mathcal{R}(A) \cap \mathcal{V}_1 + \mathcal{R}(A) \cap \mathcal{V}_2$ because the reverse inclusion $\mathcal{R}(A) \cap \mathcal{V}_1 + \mathcal{R}(A) \cap \mathcal{V}_2 \subset \mathcal{R}(A) \cap (\mathcal{V}_1 + \mathcal{V}_2)$ is true for any linear space (we have proved that in Problem 2.1(1.3)).

Let $y \in \mathcal{R}(A) \cap (\mathcal{V}_1 + \mathcal{V}_2)$. By definition, there exists $x$ such that $Ax = y$, and there exist $v_1 \in \mathcal{V}_1, v_2 \in \mathcal{V}_2$ such that $y = v_1 + v_2$.

Since $Ax = y = v_1 + v_2$, we have $x \in A^{-1}(\mathcal{V}_1 + \mathcal{V}_2)$.

By our assumption, there exist $x_1 \in A^{-1}\mathcal{V}_1$ and $x_2 \in A^{-1}\mathcal{V}_2$ such that $x = x_1 + x_2$. Then we have $y = Ax = Ax_1 + Ax_2$, where $Ax_1 \in \mathcal{V}_1$ and $Ax_2 \in \mathcal{V}_2$. Thus, $y \in \mathcal{R}(A) \cap \mathcal{V}_1 + \mathcal{R}(A) \cap \mathcal{V}_2$, which implies $\mathcal{R}(A) \cap (\mathcal{V}_1 + \mathcal{V}_2) \subset \mathcal{R}(A) \cap \mathcal{V}_1 + \mathcal{R}(A) \cap \mathcal{V}_2$.

**"$\Leftarrow$":**

Suppose $\mathcal{R}(A) \cap (\mathcal{V}_1 + \mathcal{V}_2) = \mathcal{R}(A) \cap \mathcal{V}_1 + \mathcal{R}(A) \cap \mathcal{V}_2$. We only need to show $A^{-1}\mathcal{V}_1 + A^{-1}\mathcal{V}_2 \subset A^{-1}(\mathcal{V}_1 + \mathcal{V}_2)$, since the reverse inclusion is already established by Eq. (3).

Let $x \in A^{-1}(\mathcal{V}_1 + \mathcal{V}_2)$. By definition, this means $Ax \in \mathcal{V}_1 + \mathcal{V}_2$. Since $Ax$ is clearly in the range of $A$, we have:

$$Ax \in \mathcal{R}(A) \cap (\mathcal{V}_1 + \mathcal{V}_2)$$

By our assumption, $Ax$ can be decomposed into elements of the intersections:

$$Ax = y_1 + y_2, \quad \text{where } y_1 \in \mathcal{R}(A) \cap \mathcal{V}_1 \text{ and } y_2 \in \mathcal{R}(A) \cap \mathcal{V}_2$$

Since $y_1 \in \mathcal{R}(A)$, there exists some $x_1$ such that $Ax_1 = y_1$. Furthermore, since $y_1 \in \mathcal{V}_1$, we have $Ax_1 \in \mathcal{V}_1$, which implies $x_1 \in A^{-1}\mathcal{V}_1$. Similarly, since $y_2 \in \mathcal{R}(A)$, there exists some $x_2$ such that $Ax_2 = y_2$. Since $y_2 \in \mathcal{V}_2$, we have $Ax_2 \in \mathcal{V}_2$, which implies $x_2 \in A^{-1}\mathcal{V}_2$.

Now consider the sum $x_1 + x_2$:

$$A(x_1 + x_2) = Ax_1 + Ax_2 = y_1 + y_2 = Ax$$

This implies $A(x-(x_1+x_2))=0$, so $x-(x_1+x_2)\in\mathcal{N}(A)$. Let $n=x-(x_1+x_2)$. Then $x=x_1+x_2+n$.

Since for any subspace $\mathcal{V}$, the null space $\mathcal{N}(A)\subset A^{-1}\mathcal{V}$. Let $\tilde{x}_1=x_1+n$. Then $A\tilde{x}_1=Ax_1+0=y_1\in\mathcal{V}_1$, so $\tilde{x}_1\in A^{-1}\mathcal{V}_1$.

Finally, $x=\tilde{x}_1+x_2$, where $\tilde{x}_1\in A^{-1}\mathcal{V}_1$ and $x_2\in A^{-1}\mathcal{V}_2$. Therefore, $x\in A^{-1}\mathcal{V}_1+A^{-1}\mathcal{V}_2$.

## 3. Problem 2.5

**3.1) Prove that $A(A^{-1}\mathcal{U}) = \mathcal{U} \cap \mathcal{R}(A)$.**

**"$\subset$":**

Let $x \in A(A^{-1}\mathcal{U})$. Then there exists $y \in A^{-1}\mathcal{U}$ such that $x = Ay$. Since $y \in A^{-1}\mathcal{U}$, we have $Ay \in \mathcal{U}$, so $x \in \mathcal{U}$. Also, since $x = Ay$, we have $x \in \mathcal{R}(A)$. Therefore, we have $A(A^{-1}\mathcal{U}) \subset \mathcal{U} \cap \mathcal{R}(A)$.

**"$\supset$":**

Let $x \in \mathcal{U} \cap \mathcal{R}(A)$. Since $x \in \mathcal{R}(A)$, there exists $y$ such that $Ay = x$. Since $x \in \mathcal{U}$, we have $Ay \in \mathcal{U}$, so $y \in A^{-1}\mathcal{U}$. Therefore, we have $x = Ay \in A(A^{-1}\mathcal{U})$, which implies $\mathcal{U} \cap \mathcal{R}(A) \subset A(A^{-1}\mathcal{U})$.

Overall, we can conclude that $A(A^{-1}\mathcal{U}) = \mathcal{U} \cap \mathcal{R}(A)$.

**3.2) Prove that $A^{-1}(A\mathcal{U}) = \mathcal{U} + \mathcal{N}(A)$.**

**"$\subset$":**

Let $x \in A^{-1}(A\mathcal{U})$. Then there exists $u \in \mathcal{U}$ such that $Ax = Au$. This implies $A(x - u) = 0$, so $x - u \in \mathcal{N}(A)$. Therefore, we have $x = u + (x - u)$, where $u \in \mathcal{U}$ and $x - u \in \mathcal{N}(A)$, which implies that $x \in \mathcal{U} + \mathcal{N}(A)$.

**"$\supset$":**

Let $x \in \mathcal{U} + \mathcal{N}(A)$. Then there exist $u \in \mathcal{U}$ and $n \in \mathcal{N}(A)$ such that $x = u + n$. Since $n \in \mathcal{N}(A)$, we have $An = 0$, so $Ax = Au + An = Au$. Since $u \in \mathcal{U}$, we have $Au \in A\mathcal{U}$, so $Ax \in A\mathcal{U}$. Therefore, we have $x \in A^{-1}(A\mathcal{U})$.

Overall, we can conclude that $A^{-1}(A\mathcal{U}) = \mathcal{U} + \mathcal{N}(A)$.

## 4. Problem 2.6

1. **Prove that $\mathcal{R}(A) = \mathcal{N}(A^*)^\perp$**

   This is equivalent to proving $\mathcal{R}(A)^\perp = \mathcal{N}(A^*)$.

   Let $y \in \mathcal{R}(A)^\perp$, then for all $x$ we have $\langle Ax, y\rangle = 0$. Using the definition of the adjoint:

   $$\langle Ax, y\rangle = \langle x, A^*y\rangle = 0$$

   Since this holds for all $x$, we must have $A^*y = 0$, so $y \in \mathcal{N}(A^*)$. Conversely, if $y \in \mathcal{N}(A^*)$, then for all $x$ we have $\langle Ax, y\rangle = \langle x, A^*y\rangle = 0$, so $y \in \mathcal{R}(A)^\perp$.

2. (a) **Prove that $\mathcal{N}(A) = \mathcal{N}(A^*A)$**

   ($\subset$) If $x \in \mathcal{N}(A)$, then $Ax = 0 \implies A^*Ax = 0 \implies x \in \mathcal{N}(A^*A)$.

   ($\supset$) If $x \in \mathcal{N}(A^*A)$, then $A^*Ax = 0$. Multiply by $x^*$:

   $$x^*A^*Ax = 0 \implies \langle Ax, Ax\rangle = 0 \implies \|Ax\|^2 = 0 \implies Ax = 0$$

   Therefore $x \in \mathcal{N}(A)$.

   (b) **Prove that $\mathcal{R}(A) = \mathcal{R}(AA^*)$**

   Apply the result from part (a) to the matrix $A^*$. We have $\mathcal{N}(A^*) = \mathcal{N}((A^*)^*A^*) = \mathcal{N}(AA^*)$. Taking orthogonal complements:

   $$\mathcal{N}(A^*)^\perp = \mathcal{N}(AA^*)^\perp$$

   Using the identity $\mathcal{R}(M) = \mathcal{N}(M^*)^\perp$ and the fact that $(AA^*)^* = AA^*$:

   $$\mathcal{R}(A) = \mathcal{R}(AA^*)$$

   **Prove that $\mathcal{R}(A) \subset \mathcal{R}(B)$ if and only if $\mathcal{N}(B^*) \subset \mathcal{N}(A^*)$**

   ($\Rightarrow$) If $\mathcal{R}(A) \subset \mathcal{R}(B)$, then $\mathcal{R}(A)^\perp \supset \mathcal{R}(B)^\perp$. Using the identity $\mathcal{R}(M) = \mathcal{N}(M^*)^\perp$:

   $$\mathcal{N}(A^*) \supset \mathcal{N}(B^*)$$

   ($\Leftarrow$) If $\mathcal{N}(B^*) \subset \mathcal{N}(A^*)$, then $\mathcal{N}(B^*)^\perp \supset \mathcal{N}(A^*)^\perp$. Using the identity $\mathcal{R}(M) = \mathcal{N}(M^*)^\perp$:

   $$\mathcal{R}(B) \supset \mathcal{R}(A)$$

## 5. Problem 2.13

### 5.1) Prove that the Hölder p-norms are indeed norms in $\mathbb{F}^n$.

For $1 \leq p < \infty$, the $p$-norm is defined as $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p}$. For $p = \infty$, $\|x\|_\infty = \max_{1\leq i\leq n} |x_i|$.

- (positivity): Since $|x_i|^p \geq 0$ for all $i$, we have $\|x\|_p \geq 0$. Moreover, $\|x\|_p = 0$ if and only if $|x_i|^p = 0$ for all $i$, which happens if and only if $x_i = 0$ for all $i$, i.e., $x = 0$.
  Therefore, $\|x\|_p > 0$ for all $x \neq 0$ is true.
- (homogeneity): For any scalar $\alpha$ and vector $x$, we have:

$$\|\alpha x\|_p = \left(\sum_{i=1}^n |\alpha x_i|^p\right)^{1/p} = \left(|\alpha|^p \sum_{i=1}^n |x_i|^p\right)^{1/p} = |\alpha| \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} = |\alpha| \|x\|_p$$

- (triangle inequality): For $1 \leq p < \infty$, the triangle inequality follows from Minkowski's inequality:

$$\left(\sum_{i=1}^n |x_i + y_i|^p\right)^{1/p} \leq \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^n |y_i|^p\right)^{1/p}$$

  For $p = \infty$, we have:

$$\|x + y\|_\infty = \max_{1\leq i\leq n} |x_i + y_i| \leq \max_{1\leq i\leq n} (|x_i| + |y_i|) \leq \max_{1\leq i\leq n} |x_i| + \max_{1\leq i\leq n} |y_i| = \|x\|_\infty + \|y\|_\infty$$

### 5.2) Prove that the Ky Fan $k$-norms are indeed norms in $\mathbb{F}^n$.

Let $k$ be an integer such that $1 \leq k \leq n$. The Ky Fan $k$-norm is defined as:

$$\|x\|_{(k)} = \sum_{j=1}^k |x|_j^\downarrow$$

where $|x|_1^\downarrow \geq |x|_2^\downarrow \geq \cdots \geq |x|_n^\downarrow$ denotes the components of $x$ sorted by absolute value in descending order.

- (positivity): Since $|x|_j^\downarrow \geq 0$ for all $j$, we have $\|x\|_{(k)} \geq 0$. Moreover, $\|x\|_{(k)} = 0$ if and only if $|x|_j^\downarrow = 0$ for all $j$, which happens if and only if $x_j = 0$ for all $j$, i.e., $x = 0$.
  Therefore, $\|x\|_{(k)} > 0$ for all $x \neq 0$ is true.
- (homogeneity): For any scalar $\alpha$ and vector $x$, we have:

$$\|\alpha x\|_{(k)} = \sum_{j=1}^k |\alpha x|_j^\downarrow = |\alpha| \sum_{j=1}^k |x|_j^\downarrow = |\alpha| \|x\|_{(k)}$$

- (triangle inequality): Let $x, y \in \mathbb{F}^n$. We want to show that $\|x + y\|_{(k)} \leq \|x\|_{(k)} + \|y\|_{(k)}$. Let $|x|_1^\downarrow \geq |x|_2^\downarrow \geq \cdots \geq |x|_n^\downarrow$ and $|y|_1^\downarrow \geq |y|_2^\downarrow \geq \cdots \geq |y|_n^\downarrow$ be the sorted absolute values of $x$ and $y$, respectively. By the triangle inequality for absolute values, we have:

$$|x_i + y_i| \leq |x_i| + |y_i|$$

Sorting the LHS in descending order gives us $|x+y|_1^{\downarrow} \geq |x+y|_2^{\downarrow} \geq \cdots \geq |x+y|_n^{\downarrow}$. Therefore:

$$\|x+y\|_{(k)} = \sum_{j=1}^{k} |x+y|_j^{\downarrow} \leq \sum_{j=1}^{k} (|x|_j^{\downarrow} + |y|_j^{\downarrow}) = \sum_{j=1}^{k} |x|_j^{\downarrow} + \sum_{j=1}^{k} |y|_j^{\downarrow} = \|x\|_{(k)} + \|y\|_{(k)}$$